

Algebra: Chapter 0

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1 Naive set theory

Exercise 1.1. Locate a discussion of Russell's Paradox and understand it.

Solution 1.1. Russell's paradox works by assuming that a set of all sets exists (nothing in naive set theory suggests it could not) and then constructs a subset S which is the set of all sets that are not members of themselves. Now all sets are either in S or not (because all sets are contained in the set of all sets). Now consider S itself. If we assume $S \in S$, then this contradicts that S is not a member of itself. However if $S \notin S$ then S satisfies the property of S and is a member of S . Clearly this shows that our original naive assumption that a set of all sets exists is false.

Exercise 1.2. Prove that if $\sim$ is a relation on a set S , then the corresponding family $\mathcal{P}_\sim$ is indeed a partition of S : that is, its elements are nonempty, disjoint, and their union is S .

Solution 1.2. We consider each assertion in turn. First to show nonemptiness, all elements of $\mathcal{P}_\sim$, the equivalence classes $[a]_\sim$, must contain a as $a \sim a$ by reflexivity. To show disjointness, suppose $\exists c \in [a]_\sim \cap [b]_\sim$, where $a \not\sim b$. Then by definition we have $c \sim a$ and $c \sim b$. However by transitivity and symmetry we get that $a \sim b$ contradicting our initial assumption (hence they are disjoint). Now consider the final point; $\forall a \in S$ we have $a \sim a$ so $\forall a \in S \exists [a]_\sim \in \mathcal{P}_\sim$. All elements of S are in an equivalence class, so the union is S .

Exercise 1.3. Given a partition $\mathcal{P}$ on a set S , show how to define a relation $\sim$ on S such that $\mathcal{P}$ is the corresponding partition.

Solution 1.3. Define $\sim$ as below,

$$\sim = \{(a, b) \in S \times S \mid \exists c \in \mathcal{P} \text{ s.t. } a \in c \text{ and } b \in c\} \quad (1.1)$$

then all the necessary properties follow.

Exercise 1.4. How many different equivalence relations may be defined on the set $\{1, 2, 3\}$

Solution 1.4. This amounts to partitions of 3, of which there are 5:

$$\begin{aligned} &\{1\}, \{2\}, \{3\} \\ &\{1, 2\}, \{3\} \\ &\{2, 3\}, \{1\} \\ &\{1, 3\}, \{2\} \\ &\{1, 2, 3\} \end{aligned}$$

Exercise 1.5. Give an example of a relation that is reflexive and symmetric but not transitive. What happens if you attempt to use this relation to define a partition on the set? (Hint: thinking about the second question will help you answer the first one.)

Solution 1.5. A simple finite example. For $S = \{a, b, c\}$, let

$$\sim = \{(a, a), (b, b), (c, c), (a, b), (b, a), (b, c), (c, b)\} \quad (1.2)$$

So $\sim$ is reflexive and symmetric but not transitive (as $a \sim c$). Suppose we try and define a partition on S using $\sim$. Now we have the following "equivalence classes":

$$[a]_{\sim} = \{a, b\} \quad (1.3)$$

$$[c]_{\sim} = \{b, c\} \quad (1.4)$$

$$[b]_{\sim} = \{a, b, c\} \quad (1.5)$$

So the union is S but the classes are not disjoint.

Exercise 1.6. Define a relation $\sim$ on the set $\mathbb{R}$ of real numbers by setting $a \sim b \iff b - a \in \mathbb{Z}$. Prove that this is an equivalence relation, and find a "compelling" description for $\mathbb{R}/\sim$. Do the same for the relation $\approx$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by declaring $(a_1, a_2) \approx (b_1, b_2) \iff b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$.

Solution 1.6. We have reflexivity from $a - a = 0 \in \mathbb{Z}$. Symmetry follows from $a \sim b \implies b - a = k \in \mathbb{Z}$ and $a - b = -k \in \mathbb{Z}$ so $b \sim a$. Now if $a \sim b$ and $b \sim c$ then $b - a = k_1$ and $c - b = k_2$ with $k_1, k_2 \in \mathbb{Z}$. We have then $c - a = c - b + b - a = k_2 + k_1 \in \mathbb{Z}$. Hence $a \sim c$. Referring to the decimal expansion of real numbers (among other things) gives us that $\mathbb{R}/\sim$ is represented by $[0, 1)$ (notwithstanding translation by some integers). Doing the same calculations componentwise gives $\mathbb{R} \times \mathbb{R}/\approx$ is represented similarly by $[0, 1) \times [0, 1)$. Note that if we glue the open end(s) to the closed ends of the intervals we get a circle and torus respectively (implicitly moving in the ambient $\mathbb{R}^2$ or $\mathbb{R}^3$).